Client Churn Risk & Revenue Forecasting for Managed IT Services

CountryEdu / Abhishek & Company — Data Analyst Case Study

Synthetic implementation. Production data was not supplied; therefore every record and numerical result in this project is synthetic.

1. Executive Summary

The solution converts 10 million transactions into client-month health features for 200 enterprise clients, predicts three-month churn risk, ranks clients by expected revenue loss and forecasts revenue for the next 12 months. The selected model is Logistic Regression, with ROC-AUC 0.896 and PR-AUC 0.760. Revenue forecast backtest MAPE is 1.15%.

2. Business Definitions

Churn target: a client is classified as positive for the three-month horizon using observable customer-health signals available at the observation date. Retention priority equals churn probability × annualized client revenue. The 5% churn-reduction request is interpreted as a relative reduction: 18% × 95% = 17.1% target churn.

3. Dataset

Dimension	Value
Enterprise clients	200
Transaction records	10,000,000
Financial missingness	12%
Baseline churn	18%
Prediction horizon	3 months
Forecast horizon	12 months

Financial missing values are retained with explicit missingness indicators. Numerical model predictors are imputed inside the training pipeline; clients are not removed solely because a financial field is missing.

4. Feature Engineering

Features cover monthly revenue, revenue growth, payment delays, support-ticket volume, incidents, escalations, SLA compliance, security events, cloud usage, CSAT, CSAT trend, contract timing, service count and three-month trends.

5. Churn Model Evaluation

Model	ROC-AUC	PR-AUC	Precision	Recall	F1
Logistic Regression	0.896	0.760	0.511	0.818	0.629
XGBoost	0.889	0.746	0.733	0.555	0.631

Logistic Regression is selected using PR-AUC first and ROC-AUC second because churn is an imbalanced classification problem.

6. Highest-Risk Clients

Client	Churn probability	Risk	Annual revenue	Expected loss
C002	100.0%	Critical	3,396,272	3,396,006
C097	99.9%	Critical	2,656,236	2,654,272
C114	96.5%	Critical	2,241,547	2,163,819
C079	97.9%	Critical	2,139,233	2,093,497
C050	95.5%	Critical	1,934,660	1,846,873
C185	99.3%	Critical	1,711,102	1,698,749
C087	96.2%	Critical	1,408,913	1,355,423
C036	81.9%	Critical	1,629,977	1,334,531
C102	96.7%	Critical	1,350,532	1,306,523
C004	78.0%	Critical	1,673,829	1,305,095

The top 20 synthetic clients represent approximately 30,640,404 of expected revenue loss.

7. Revenue Forecast

Holt-Winters exponential smoothing uses additive trend and 12-month seasonality. The final six historical months are held out for backtesting. MAPE is 1.15%. The 12-month base forecast is 159,486,763.

The churn model estimates 79,347,988 of annualized revenue exposure. The risk-adjusted planning view is 80,138,774; this is a risk scenario rather than a claim that all predicted churn will occur.

8. Retention Strategy

Critical/high-revenue-risk clients receive executive retention and renewal intervention. Other critical clients receive service recovery and account-manager review. High-risk clients receive a health check plus SLA/CSAT improvement plan. Medium and low-risk clients remain under proactive monitoring.

9. Management KPIs

Churn rate, target churn, clients at risk, expected revenue loss, retention conversion, CSAT, SLA compliance, support escalation rate, base revenue forecast and risk-adjusted revenue forecast.

10. Final Recommendation

Use churn probability to identify risk, expected revenue loss to prioritize intervention, and the risk-adjusted forecast to quantify financial exposure. The management target is 17.1% churn from an 18% baseline.